Xi Chen

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Profile

Statistics PhD candidate specializing in statistical machine learning, graph-structured data, and multivariate time series analysis. Experienced in developing predictive and unsupervised models, designing large-scale simulation studies, and building reproducible Python and R workflows for complex real-world data. Seeking a 2027 data science internship.

Education

University of California, Santa Cruz	PhD, Statistics; candidacy: 07/2026	2023–Expected 06/2028
University of California, Davis	MS, Biostatistics	2021–2023
Sun Yat-Sen University	BS, Applied Mathematics	2014–2018

Technical Skills

Programming & Data: Python (NumPy, pandas, PyTorch, scikit-learn), R (Rcpp), SQL (SQLite), MATLAB
Statistics & Machine Learning: statistical inference, hypothesis testing, regression, classification, time-series analysis, changepoint detection, graph-based learning, machine learning, deep learning, Bayesian statistics
Tools: Git, SSH, Shell/Linux, Jupyter, LaTeX

Data Science & Research Experience

PhD Candidate & Graduate Researcher Sep 2023–Present
University of California, Santa Cruz

Multivariate Changepoint Detection *(manuscript nearing submission)*

- Co-developed a test for coordinated mean shifts across correlated multivariate time series. The change time is shared, but each variable may shift by a different amount or direction; Implemented weighted vector U-statistics to detect changepoints near the boundary of a time series, and pre-whitening to account for serial dependence.
- Evaluated false-positive control and detection power in studies with up to 300,000 VAR simulations per setting; analyzed cardio-respiratory measurements from a CO_2 inhalation experiment. Rcpp made the adjusted simulation workload about 4× faster than an R reference in a 1,000-run benchmark.

Latent-Space Graph-Fused LASSO [*arXiv preprint*]

- Co-developed unsupervised clustering for nodes with both feature vectors and graph links. Learned compact representations of node attributes and used graph-based fusion to encourage related nodes to share clusters.
- Reached 97.0% normalized mutual information versus 83.1% for the strongest listed baseline on a 144-node simulated grid. Used shared-border links and 168 monthly temperature observations to group all 58 California counties into 10 climate regions.

GraphClustAR *(manuscript in preparation)*

- Co-developing a method to cluster graph-linked time series by similar autoregressive dynamics, making groups interpretable through their temporal behavior. Formulated graph-fused estimation, ADMM optimization, and BIC-based model selection.

Research Assistant Sep 2019–Jul 2021
Shenzhen Institutes of Advanced Technology, Chinese Academy of Sciences

- **Blood-pressure estimation:** Built supervised models to estimate blood pressure from ECG/PPG signals for potential wearable monitoring; filtered signals, segmented waves, and removed artifacts before training.
- **Arrhythmia detection:** Analyzed paired ECG/PPG recordings and co-authored a *JAHA* study; its PPG-based CNN classified six rhythms with 85.0% accuracy and 0.978 micro-AUROC on 23,384 held-out waveforms.

Selected Publications & Preprints

- Liu, Z., Zhou, B., Jiang, Z., **Chen, X.**, et al. (2022). Multiclass arrhythmia detection and classification from photoplethysmography signals using a deep convolutional neural network. *Journal of the American Heart Association*, 11(7).
- Kei, Y. L., Padilla, O. H. M., Killick, R., Wilson, J., **Chen, X.**, and Lund, R. (2025). Decoder-only clustering in attributed graphs. arXiv:2511.04859.